

NETHOLABS RESEARCH

DATA QUALITY REPORT

FUNCTIONAL BRAIN EMULATION · 67 GB ACROSS 42 RECORDINGS

● CRITICAL ● WARNING ● INFO

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CRITICAL ISSUES

Block analysis or reduce validity

3

WARNINGS

May affect data quality

2

INFO

Observations for cleanup

ISSUES IDENTIFIED

Single Subject Dominance

All 8 largest session recordings appear to be from the same participant. The 42 XDF files list 9 subject IDs, but the richest multi-modal data comes from undifferentiated `session_*` timestamped files. Between-subject variability cannot be assessed with current data.

Subject distribution:

session_* (unnamed) – 14 files, 3.1 GB, 8 modalities each
sub-P001 – 6 files, 13.8 GB (includes 5 _old versions)
sub-P002 – 2 files, 3.7 GB (includes 1 _old version)
sub-P007, P009, 090, 0903, 09031, 0999, 2003 – 1-3 files each, smaller

CRITICAL

GENERALIZABILITY

No Session Contains All Modalities

The lab stack and mobile stack have never been recorded simultaneously. No single XDF file contains all 11 modalities. This means cross-stack correlations (e.g., EEG vs. fNIRS, or DSI-24 vs. Kernel Flow) cannot be computed from any existing recording.

Lab sessions contain: Gazeport + KernelFlow (x3) + StretchSense + Microphone
Mobile sessions contain: DSI24 + TobiiGlasses (x2) + StretchSense
Missing overlap: EEG never co-recorded with fNIRS
Missing overlap: EMG (Delsys) not present in any exported session

CRITICAL

CROSS-MODAL ANALYSIS

Kernel Flow Sample Rate Mismatch

All three KernelFlow LSL streams (Red, IR, Quality) report a nominal rate of 5 Hz in their stream metadata, but pyxdf measured effective rates of 8–13 Hz across sessions. This 2–3x discrepancy means LSL timestamp correction may be miscalculating, causing synchronization drift with other streams.

Reported rate: 5.0000 Hz
Measured rates across sessions:
KernelFlow_Red – 8.08 – 13.14 Hz
KernelFlow_IR – 8.32 – 13.13 Hz
KernelFlow_Quality – 0.99 – 13.62 Hz
Quality stream in one session dropped to ~1 Hz

CRITICAL

SYNCHRONIZATION

Short Usable Overlap Windows

Many sessions have limited time windows where all streams are simultaneously active. Streams start and stop at different times, reducing the usable overlapping segment. One session had only 37 seconds of multi-stream overlap.

```
session_20260306_171345.xdf: 37s overlap (file is 22.5 MB)
session_20260227_145319.xdf: <5s usable (file is 241 KB)
Best sessions (Mar 5-6): 120s+ overlap available
Effective yield: ~877s total across 8 sessions (vs. potential minutes of raw data)
```

WARNING

USABLE DURATION

Gazepoint Rate Anomaly

In one session, the Gazepoint stream was measured at 147 Hz instead of the expected 60 Hz. This may indicate duplicate samples, a misconfigured Gazepoint Control setting, or two Gazepoint streams being merged.

```
Expected rate: 60.0000 Hz
Measured rate: 147.1897 Hz
Session: One of the Feb 27 - Mar 6 recordings
Other sessions: Normal ~60 Hz
```

WARNING

DATA INTEGRITY

_old File Accumulation

13 XDF files carry _old1 through _old5 suffixes, suggesting LabRecorder overwrites or manual re-recordings. These inflate the total data footprint by ~15 GB but may contain partial, corrupted, or superseded data.

```
Examples:
sub-P001_..._eeg_old1.xdf - 4.6 GB
sub-P001_..._eeg_old2.xdf - 7.1 GB
sub-P001_..._eeg_old3.xdf - 1.3 GB
sub-P001_..._eeg_old4.xdf - 421 MB
sub-P001_..._eeg_old5.xdf - 176 MB
+ 8 more _old files across other subjects
```

WARNING

STORAGE / CLARITY

Aborted / Minimal Recordings

Several XDF files are extremely small (<1 MB), likely from aborted recording attempts or quick equipment tests. These contain insufficient data for any analysis.

```
Files under 1 MB:
sub-P009_ses-S009_...eeg.xdf - 56 KB
```

session_20260227_145319.xdf — 241 KB

session_20260305_150657.xdf — 44 KB

sub-P001_ses-S001_...eeg.xdf — 301 KB

sub-090_ses-090_...xdf — 426 KB

INFO

CLEANUP

Inconsistent BIDS Naming

Subject and session identifiers use mixed conventions. Some use zero-padded numeric IDs (sub-P001, ses-S001), others use bare numbers (sub-090, ses-1), and many have no subject ID at all (session_YYYYMMDD_HHMMSS). This complicates automated analysis pipelines.

Conventions observed:

sub-P001_ses-S001_task-Default_run-001_eeg.xdf — structured

sub-090_ses-010_task-Baseline_run-001_.xdf — different pattern

sub-2003_ses-1_task-Baseline_run-001_.xdf — no zero-padding

session_20260306_091530.xdf — no subject ID

INFO

ORGANIZATION

MODALITY COVERAGE PER RECORDING TYPE

SESSION TYPE	EEG	FNIRS OXY	FNIRS DEOXY	FNIRS QA	GAZE (LAB)	GAZE (MOBILE)	EMG	HAND/BODY
Lab sessions (session_*)	—	✓	✓	✓	✓	—	—	✓
sub-P001 (BIDS)	?	✓	✓	✓	✓	—	—	✓
Mobile sessions	✓	—	—	—	—	✓	—	✓
Full coverage target	✓	✓	✓	✓	✓	✓	✓	✓

FILE SIZE DISTRIBUTION (TOP 15)

P001_old2		7.1 GB	_OLD
P001_old1		4.6 GB	_OLD
P002_old1		3.7 GB	_OLD
P007_ses-S007		1.7 GB	GOOD
090_ses-030		1.5 GB	GOOD

P001_old3		1.3 GB	_OLD
ses_0306_0915		967 MB	GOOD
09031_ses-1		732 MB	GOOD
ses_0305_2049		643 MB	GOOD
ses_0302_1702		550 MB	GOOD
ses_0305_2107		508 MB	GOOD
P001_old4		421 MB	_OLD
ses_0306_1811		218 MB	GOOD
ses_0302_1329		191 MB	GOOD
P001_old5		176 MB	_OLD

RECOMMENDATIONS

- 01 Run a combined lab + mobile session. Connect DSI-24, Kernel Flow, Gazepoint, StretchSense, and Delsys simultaneously into a single LabRecorder instance. This is the highest-priority gap — without it, cross-modal EEG-fNIRS analysis is impossible.
- 02 Fix Kernel Flow stream metadata. Update the nominal_srate in kernel_flowIsl.py to match the actual output rate (~10-13 Hz), or configure Kortex to output at a stable declared rate. This will fix LSL timestamp correction.
- 03 Record at least 3 more subjects with the full multi-modal stack to establish between-subject baselines. Target 5+ minute continuous recordings per session for meaningful spectral and correlation analysis.
- 04 Standardize BIDS naming. Use consistent zero-padded subject/session IDs across all recordings. Move session_* files into the sub-XXX folder structure. Archive or delete _old files after verification.
- 05 Add pre-recording checklist to start_all.bat that verifies all streams are active and at expected rates before LabRecorder begins. This will prevent aborted/empty recordings and catch rate anomalies early.